The Ising Model:
Physics and Project Guide for 3B

Maryam Ahsan

Mathematical Methods for Physicists II

Project Guide

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This is a collection of learning notes accompanying the Monte Carlo part of my Mathematical Methods for Physicists II materials and the later guided Ising-model project, notebook 3B. The Ising project is a new learning extension rather than an earlier coursework submission. These notes introduce its statistical physics before walking through the finite-lattice simulation. General Monte Carlo and Markov-chain methods are covered in the shared foundations guide and referenced here.

1 Reading route and the physical question

Read the shared *Monte Carlo Methods: Concepts and Computational Foundations* guide for random sampling, Markov chains, Metropolis acceptance, warmup, and correlation-aware uncertainty. Those general computational ideas are not rederived here. This guide explains the new Ising physics, then follows each section of notebook 3B and connects that physics to the algorithm.

The project asks how interactions and temperature influence order, energy, and fluctuations in a finite spin lattice. It is a later guided learning project, not a claim of earlier coursework. A simulated lattice is an idealized model of interacting magnetic degrees of freedom, not a microscopic movie of a real material.

2 What an Ising spin represents

An Ising spin is a two-valued variable $s_i = +1$ or -1. It represents a magnetic orientation restricted to two possibilities. The model deliberately omits continuous spin directions and precession. A *configuration* is a complete assignment of spins to every site, not one spin alone.

For a square lattice of side L, the number of spins is $N = L^2$ and the number of possible configurations is 2^N. A 20×20 lattice has 2^{400} configurations, so direct enumeration is infeasible for this project. Sampling makes averages tractable without visiting every configuration.

Each site interacts with its four nearest neighbors. The arrangement is two dimensional, and the interactions are local even though correlated domains can span many sites. A random initial configuration has roughly equal numbers of plus and minus spins, but it is not automatically a thermal-equilibrium configuration at the chosen temperature.

3 Interaction energy, field, and bond counting

The nearest-neighbor model is

$$E = -J \sum_{\langle i,j \rangle} s_i s_j - H \sum_i s_i.$$

Here each neighboring pair is counted once. The field coefficient H has energy units and includes the magnetic-moment factor in this simplified convention. The notebook sets $H = 0$ and $J = 1$.

For $J > 0$ the interaction is ferromagnetic: aligned neighbors have bond energy $-J$ and opposite neighbors $+J$. Alignment lowers energy. An antiferromagnetic coupling $J < 0$ would favor opposite neighbors and is not simulated here.

On the periodic square lattice each site has four bonds, but each bond belongs to two sites. Thus there are $4N/2 = 2N$ distinct bonds. The fully aligned ground states, all-plus and all-minus, both have

$$E_0 = -2JN, \qquad e_0 = E_0/N = -2J.$$

Counting all four neighbors at every site without dividing by two would double this energy incorrectly. The code avoids that by counting only the right and lower bond of each site.

4 Periodic boundaries and domains

Periodic boundaries identify opposite edges: the neighbor to the right of the last column is in column zero. The analogous wrapping applies vertically. The modulo operation implements this adjacency; it avoids lower coordination at open edges and represents a finite periodic system, not literally an infinite lattice.

A domain is a patch of mostly aligned spins. Domain walls contain opposite-spin bonds, each costing $2J$ relative to alignment. Flipping an isolated spin in an all-up lattice creates four unfavorable bonds and costs $8J$. This connects the local update formula to physical defects.

At low temperature such costly defects are suppressed. At higher temperature they are more frequent and domains can rearrange. A snapshot is one fluctuating configuration; a large patch in one image is not, on its own, proof of bulk magnetic order.

5 Temperature, Boltzmann weights, and entropy

5.1 The canonical ensemble

The lattice exchanges energy with a thermal reservoir at temperature T, so its energy is not conserved during Monte Carlo sampling. Its equilibrium probability is

$$P(c) = \frac{e^{-E(c)/(k_B T)}}{Z}, \qquad Z = \sum_c e^{-E(c)/(k_B T)}.$$

The partition function normalizes the distribution. An equilibrium observable is

$$\langle A \rangle = \sum_c A(c) P(c).$$

The foundations guide explains how a Markov chain generates these nonuniform weights without evaluating Z.

Setting $k_B = 1$ measures temperature in energy units. With $J = 1$, the displayed temperatures are numerically $k_B T/J$. Reintroducing units requires this ratio; the numbers 1.5 and 3.5 are not specified in kelvin.

5.2 Why energy minimization is not enough

Low-energy configurations receive higher weight individually, but there can be vastly more disordered configurations than ordered ones. If $g(E)$ denotes the number of configurations at an energy, the probability of that energy is proportional to $g(E)e^{-\beta E}$. The entropy associated with multiplicity is $S = k_B \ln g$.

The competition summarized by free energy $F = U - TS$ explains why order favored energetically can disappear at higher temperature. The sampling must allow occasional energy increases to capture this competition; accepting only downhill flips would not reproduce finite-temperature equilibrium.

A Monte Carlo sequence provides configurations distributed according to this equilibrium model when adequately equilibrated and mixed. Sweeps are algorithmic sampling steps, not calibrated physical seconds. Energy fluctuation is physical in the canonical ensemble, unlike numerical energy drift in the conservative orbit project 2C.

6 Magnetization and the meaning of order

Define total magnetization $M = \sum_i s_i$ and magnetization per spin $m = M/N$. All-up gives $m = 1$ and all-down $m = -1$. A balanced configuration has m close to zero. The field couples to this signed magnetization, while $|m|$ measures alignment without selecting a direction.

At zero field every configuration and its globally reversed partner have the same energy. A fully sampled finite equilibrium ensemble therefore has exact $\langle M \rangle = 0$, even at low temperature when individual configurations are strongly ordered. For this reason the notebook reports $\langle |m| \rangle$ as its order diagnostic.

Taking the absolute value *before* averaging is essential: $\langle |M| \rangle$ differs from $|\langle M \rangle|$. In a chain visiting both ordered signs, the latter can vanish while the former remains large. A short chain can remain in one sign; apparent nonzero signed magnetization is then not proof of full equilibrium symmetry breaking.

For an infinite system, spontaneous magnetization is defined with the thermodynamic limit taken before removing an infinitesimal symmetry-selecting field. This differs from the symmetric finite zero-field ensemble. The notebook focuses on finite-lattice observations, not that limiting construction.

7 Low-temperature and high-temperature benchmarks

At very low temperature the ferromagnetic lattice is almost aligned, so $\langle |m| \rangle$ is near one and $\langle E \rangle/N$ near $-2J$. Defects create small departures. At infinite temperature all configurations become equally likely, independent spins have zero mean, and the mean bond product vanishes, giving mean energy per spin zero.

For independent zero-field spins, $\mathrm{Var}(M) = N$, so typical m is of order $N^{-1/2} = 1/L$. For a large independent-spin lattice,

$$\langle |m| \rangle \approx \sqrt{\frac{2}{\pi N}}.$$

Thus a finite disordered lattice can have nonzero absolute magnetization. At the high-temperature limit the approximate values are 0.0665 for $L = 12$ and 0.0399 for $L = 20$. At the finite temperatures in the notebook, residual interactions change these values; they are limiting checks rather than exact predictions for the displayed curves.

8 The transition and a finite-lattice crossover

For the infinite isotropic square-lattice model with zero field,

$$T_c = \frac{2J}{k_B \ln(1 + \sqrt{2})}.$$

With the notebook's normalization, $T_c \approx 2.269$. Below this temperature the infinite model supports ordered phases; above it the spontaneous magnetization vanishes. Near the transition, correlated regions become large and thermal fluctuations become especially important.

A finite lattice has a finite partition sum and no thermodynamic singularity at positive temperature. Its temperature curves are rounded. Peaks in finite-lattice fluctuation measures can shift with system size. A dashed T_c line in these plots is an externally known infinite-system benchmark, not a value inferred from the simulation.

Local Metropolis updates can decorrelate slowly near criticality: critical slowing down. The temperature close to 2.3 is therefore particularly challenging for the project's short runs. Increasing lattice size can increase both physical correlations and sampling difficulty.

9 Fluctuations and thermodynamic response

9.1 Heat capacity from energy fluctuations

For a temperature-independent Hamiltonian, differentiation of the canonical average gives

$$\frac{\partial \langle E \rangle}{\partial \beta} = -(\langle E^2 \rangle - \langle E \rangle^2).$$

Since $\partial\beta/\partial T = -1/(k_B T^2)$,

$$\frac{C_V}{N} = \frac{\mathrm{Var}(E)}{N k_B T^2}.$$

For $k_B = 1$, this is the code's energy-variance formula. Heat capacity measures how mean energy responds to temperature; canonical energy fluctuations encode that response. Calculate the variance of *total* energy, then divide by N. The variance of energy per spin would need an extra factor N to give the same quantity.

9.2 Susceptibility from magnetization fluctuations

With the field term $-HM$, susceptibility per spin in this convention is

$$\chi = \frac{1}{N} \frac{\partial \langle M \rangle}{\partial H} = \frac{\mathrm{Var}(M)}{N k_B T}.$$

Differentiating the Boltzmann weights with respect to H gives the covariance with M, yielding this variance. It measures response to a small field, not simply the absolute alignment of the lattice. Setting $H = 0$ for the simulation does not make the derivative-based response zero.

Both formulas concern equilibrium averages. A finite correlated record only approximates them, and a record trapped in one sector can miss part of the equilibrium fluctuations. Choosing more temperatures does not repair this sampling limitation.

9.3 The important finite zero-field susceptibility caveat

The full finite zero-field ensemble has $\langle M \rangle = 0$. At very low positive temperature its two nearly aligned sectors have $M \approx \pm N$. Thus $\langle M^2 \rangle \approx N^2$ and the full ensemble susceptibility behaves as $\chi \approx N/T$ for $k_B = 1$.

A short chain initialized all-up often samples only the positive sector. Its empirical $\langle M \rangle$ is then near N, and subtracting the empirical mean squared removes the large between-sector contribution. The resulting small variance describes fluctuations in the sampled sector; it does not estimate the fully sampled symmetric susceptibility reliably.

Accordingly, the full finite zero-field susceptibility need not display only a near-T_c peak. A small low-temperature point in the project's short run is not evidence that the true full-ensemble quantity is small. Replacing $\langle M \rangle^2$ with $\langle |M| \rangle^2$ gives a different order-fluctuation diagnostic; it must not be silently identified with the same finite-system linear response.

This project retains the correct stated formula and discusses its finite-run limitations. It does not claim quantitatively converged thermodynamic susceptibilities.

10 Walkthrough 1: create a lattice and compute observables

The opening example makes a random 12×12 grid with integer spins plus or minus one. An ordered option returns all ones. NumPy creates the configurations; Python's separate random generator later selects Metropolis proposals, so both are seeded once for reproducibility.

In the energy function, loop over rows and columns. For each site use column $(j + 1)$ mod L for its right neighbor and row $(i + 1)$ mod L for its lower neighbor. Add $-Js_{ij}(s_{i,j+1} + s_{i+1,j})$. This counts each bond once, including the periodic boundary bonds.

The magnetization function simply sums the signed entries. For all-up $L = 12$, check $N = 144$, $E = -288$ with $J = 1$, and $M/N = 1$. The notebook asserts the energy identity. A random configuration can have nonzero m because it is one finite sample.

The lattice sizes used here are large enough for the usual four-neighbor periodic geometry. Tiny $L = 1$ or $L = 2$ periodic lattices require special care about coincident neighbor labels and bond conventions; they are outside the project settings.

11 Walkthrough 2: derive and implement a local flip

11.1 Derive the energy change before sampling

Let the selected spin be s and its four-neighbor sum be S. Its affected bond contribution changes from $-JsS$ to $+JsS$, so

$$\Delta E = 2JsS.$$

For zero field S is one of $-4, -2, 0, 2, 4$, and the possible changes are $-8J, -4J, 0, 4J, 8J$. No distant bonds change. Magnetization changes by $-2s$ because one contribution goes from s to $-s$.

For example, flipping an up spin surrounded by up spins has $\Delta E = 8J$. Its acceptance is $e^{-8/T}$ when $J = k_B = 1$, about 0.0048 at $T = 1.5$ and 0.102 at $T = 3.5$. This concrete defect is more easily created at higher temperature.

11.2 Apply the shared Metropolis rule

Choose a row and column uniformly. Evaluate the neighbor sum in the *current* grid, calculate ΔE, and apply the acceptance rule from the foundations. If accepted, negate the spin, add ΔE to stored energy, subtract $2s$ from stored magnetization, and increment the acceptance count. If rejected, leave these quantities unchanged.

One sweep means L^2 proposals with replacement. It does not ensure that each site is selected exactly once or that each proposal is accepted. The implementation is sequential: an accepted flip changes the neighbors

seen by later proposals in the same sweep. Simultaneously flipping arbitrary adjacent sites using old neighbor values would not implement this algorithm.

The notebook checks 30 sweeps at $T = 2.3$ and verifies that incremental energy and magnetization agree with fresh direct sums. This is an important bookkeeping test; agreement does not alone establish equilibration or correct uncertainty estimates.

12 Walkthrough 3: examine temperature snapshots

For each $T = 1.5, 2.3, 3.5$, start a separate all-up 24×24 grid and perform 350 sweeps. Display its final spin configuration and signed magnetization per spin.

Expect stronger alignment at low temperature and more disorder at higher temperature. The middle case lies near the infinite-lattice benchmark, where large patches and slow rearrangements may occur. One picture cannot locate the transition, certify equilibrium, or characterize all possible configurations.

The snapshots use $L = 24$, while the main temperature averages use $L = 20$. Do not interpret the two displays as measurements on the same lattice. Starting all-up and waiting a fixed number of sweeps can leave residual initial-state dependence, particularly near the transition.

13 Walkthrough 4: warm up, then measure

The function `measure_temperature` starts an ordered grid, computes its initial energy and magnetization, and discards 300 warmup sweeps. It then performs 600 measurement sweeps, recording M and E every tenth sweep. This produces 60 records per temperature.

Compute $\langle |M| \rangle / N$ by taking the absolute value of each record *before* averaging. Compute mean energy by averaging total-energy records and dividing by N. Use the total-magnetization and total-energy moments in the two fluctuation formulas. Their moment subtraction is a plug-in estimate of thermal variance from a finite record, not a proof of independently sampled fluctuations.

Acceptance fraction is the number of accepted proposals across all 600 measurement sweeps divided by $N \times 600$. Warmup acceptances are excluded. A configuration is recorded on schedule even if recent proposals were rejected. Selecting only accepted configurations would bias the equilibrium sampling.

Near T_c, 300 warmup sweeps and 60 recorded observations can be inadequate. Recording every ten sweeps reduces the record size and may reduce correlation but does not guarantee independence. The foundations explain correlation time, effective sample size, and blocking; the project does not compute them.

14 Walkthrough 5: read the temperature and fluctuation plots

The main 20×20 averages use

$$T = 1.5, 1.8, 2.0, 2.15, 2.3, 2.45, 2.6, 2.9, 3.5.$$

The alignment plot generally drops as temperature rises. Mean energy generally rises from near -2 toward zero as unfavorable bonds become more common. Acceptance generally increases because costly proposals are less strongly suppressed, but acceptance is a property of the update algorithm, not an order parameter or equilibrium certificate.

The fluctuation plots display the short-run susceptibility and heat-capacity estimates. Correlated rearrangements can give enhanced energy fluctuations near the transition; finite size and coarse temperature sampling round and shift visible peaks. The susceptibility requires the sector-sampling qualification developed above. Neither plot has correlation-aware error bars, and neither determines an accurate T_c from its highest point.

Interpret the known critical-temperature line as context, not as a fitted result. Small wiggles or unexpected peaks can reflect sampling noise and incomplete relaxation. A longer run from a different starting configuration is a meaningful check; choosing a seed solely because its picture looks expected is not validation.

15 Walkthrough 6: compare two finite sizes

The extension repeats the same averaging protocol with $L = 12$ and overlays its $\langle |m| \rangle$ curve with the $L = 20$ result. Both curves should be rounded because both systems are finite. Differences can contain actual finite-size effects, Monte Carlo noise, and different equilibration or correlation times.

The smaller lattice can have a larger high-temperature absolute magnetization simply because independent-spin fluctuations scale as $1/L$. This is not evidence that it remains spontaneously ordered above T_c. Close to the transition, correlations and finite system size complicate that simple independent-spin estimate.

Two sizes, one short run per temperature, and no error bars are a first comparison, not a finite-size-scaling analysis. Quantitative work would require longer equilibrated records, repeated or blocked uncertainty estimates, more sizes, and observables suited to extracting limiting behaviour. Those are further investigations rather than results already established by this project.

16 What an independent implementation should establish

Before interpreting temperature curves, establish the ground-state energy, the single-flip energy difference, and agreement of incremental with directly recomputed observables. Use the same field, coupling, boundary convention, and normalizations in all parts of the code.

Then ask whether the initial transient has decayed, whether all relevant sectors have been explored, and whether recorded observations remain correlated. These are sampling questions distinct from correctness of the local energy formula. Physical interpretation needs both.

The project supports a qualitative account of how thermal fluctuations disrupt alignment and how finite lattices differ from the infinite benchmark. Its limited sampling does not establish precise response coefficients or critical exponents. It remains a useful starting point because the model, local update, normalization, and limitations are explicit.

17 Book references and reading route

1. Harvey Gould and Jan Tobochnik, *Statistical and Thermal Physics: With Computer Applications*, Princeton University Press. Use its microscopic-to-macroscopic perspective, canonical ensemble, fluctuations, and computer applications for undergraduate physical intuition.
2. M. E. J. Newman and G. T. Barkema, *Monte Carlo Methods in Statistical Physics*, Oxford University Press. Consult its spin-model simulations and analysis of correlated Monte Carlo measurements. Author book page: https://websites.umich.edu/~mejn/nbook/.
3. David P. Landau and Kurt Binder, *A Guide to Monte Carlo Simulations in Statistical Physics*, Cambridge University Press. Use its treatment of phase transitions, finite-size effects, and sampling reliability as a deeper reference.

Read the shared Monte Carlo foundations for the computational machinery, then use this guide and Gould–Tobochnik for the Ising physics.